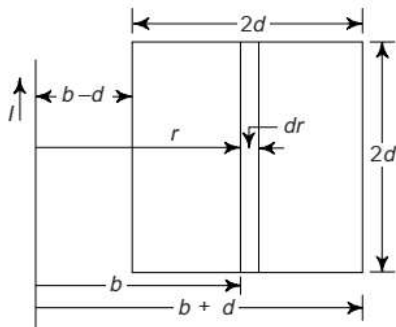


SUBJECT NAME: ELECTRIC MACHINE-I

SUBJECT CODE: PCCEEE401

ASSIGNMENT: 01

1. The no-load current of measured on the LV side of a 440/240 V, 50 Hz single-phase transformer is 4 A at 0.2 pf lagging. The voltage per turn value is 1.04. Calculate (i) maximum flux in the core, (ii) core loss, and (iii) magnetizing current.
2. Calculate the equivalent circuit parameters of a one-phase, = kVA, 400/200 V, 50 Hz transformer with the following test data: OC test: 200 V, 1.2A, 90 W (on LV side) SC test: 25 V, 12 A, 100 W (on HV side).
3. The core loss of a single-phase 440/220 V, 50 Hz transformer is 1450 W measured at its high voltage side with its low voltage being kept open. When the voltage and frequency are changed to 300 V and 34 Hz, respectively, the core loss becomes 700 W. Find values of Hysteresis and Eddy current loss of the transformer for 440 V, 50 Hz input.
4. The primary and secondary windings of a 20 kVA, 3300/220 V single-phase transformer have resistances of 9 W and 0.05 W, respectively. The total leakage reactance referred to the primary is 25 W. Find full load voltage regulation at 0.8 power factor lagging.
5. The efficiency of a 20 KVA, 2500/200 V single phase transformer at unity power factor is 98% at full load and also at half load. Determine (i) its core loss and copper loss at full load, (ii) the load at which the maximum efficiency occur and the value of maximum efficiency?
6. A square loop of side $2d$ is placed with two of its sides parallel to an infinitely long conductor carrying current I . The centre line of the square is at distance b from the conductor. Determine the expression for the total flux passing through the loop. What would be the loop flux if the loop is placed such that the conductor is normal to the plane of the loop? Does the loop flux in this case depend upon the relative location of the loop with respect to the conductor?



7. A steel ring has a mean diameter of 20 cm, a cross-section of 25 cm² and a radial air-gap of 0.8 mm cut across it. When excited by a current of 1A through a coil of 1000 turns wound on the ring core, it produces an air-gap flux of 1 mWb. Neglecting leakage and fringing, calculate (a) relative permeability of steel, and (b) total reluctance of the magnetic circuit.
8. Draw a single-layer unbifurcated winding for a 3-phase, 4-pole machine having 24 armature slots. Assume one coil-side. Clearly show the end connection if a continuous chain arrangement is used.